ISIC-2024 SLICE-3D — An Efficiency-Frontier, Single-Dataset, No-External-Data Study

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1 Abstract

We study the ISIC-2024 SLICE-3D skin-lesion task — classifying ~401,059 lesion crops extracted from 3D total-body photography as malignant or benign — under a deliberately strict regime: the **SLICE-3D dataset only**, **no external dermoscopy data**, and **no generative / synthetic positives**. The competition's top private-leaderboard solutions reached pAUC ≈ 0.173, but did so by importing external archives and ~30,000 diffusion-synthesized malignant lesions; we ban both. Because the absolute number is then out of reach by construction, we re-pose the task as a **quality-vs-cost Pareto frontier**: the best official partial-AUC above 80% TPR achievable per unit of inference cost (parameters, FLOPs, single-thread CPU latency). A hybrid pipeline — a LightGBM tabular expert over intrinsic engineered features plus a small ImageNet-pretrained image backbone, fused by a trivial rank-average — reaches **out-of-fold pAUC 0.17430** at 15.8 M params / 2.46 GFLOPs / 61 ms CPU. The recurring finding: **every increase in model complexity we tried lost.** At 393 positives, the small backbone, the zero-parameter combiner, and intrinsic tabular features are both cheaper and more accurate.

2 The task and the metric

ISIC-2024 SLICE-3D contains **401,059** lesion crops from **1,042** patients: **393 malignant** vs **400,666 benign** — a prevalence of **0.098%** (≈ 1 in 1,021). The official metric is the **partial AUC above 80% TPR** (pAUC@80), the area under the ROC restricted to the high-sensitivity band TPR ≥ 0.80; it is bounded in `[0, 0.20]` (random ≈ 0.02, perfect = 0.20). We pin **80% TPR** (not the 88% mentioned in the public metrics repo README) and compute it only via `src/cv.py`, proven numerically identical to the vendored official scorer.

3 Scope and discipline

Three locked constraints make the claim meaningful: (1) **single dataset** — ISIC-2024 SLICE-3D only, no ISIC-2019/2020, no PAD-UFES, no external dermoscopy; ImageNet-pretrained *weights* are allowed (no skin labels), external *training data* is not; (2) **no synthetic augmentation** — fabricating pathology is a label-validity hole; classical augmentation only; (3) **sacred validation** — patient-grouped, target-stratified StratifiedGroupKFold (5 folds, SEED = 42), frozen once into `data/folds.parquet`, shared by every model, with a `cv-guardian` agent holding veto power. In this competition leaky CV sent teams ~200 places down on the private split.

4 Data and EDA

The crops are variable-size square JPEGs, 61–239 px (median/mean ≈ 131/133 px), ~2.8 KB each, so 128 px training is near-native and 224 px *upsamples* (the 224 px gain is capacity/training, not extra pixels). Head/neck carries ~7× the baseline malignant rate; hue `tbp_lv_H` reaches univariate AUC 0.81; malignant lesions are ~2× larger; and the **ugly-duckling sign** is quantitatively present — 392/393 positives sit among the same patient's benign moles, at a median **88th within-patient size percentile**. Train-only leak columns (`iddx_*`, `mel_thick_mm`, `mel_mitotic_index`, `lesion_id`, `tbp_lv_dnn_lesion_confidence`) are dropped.

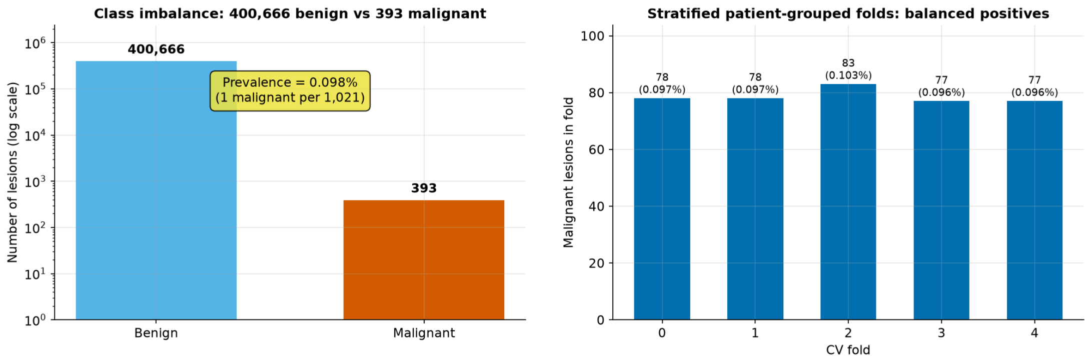

Figure 1: Class imbalance and per-fold positive counts.

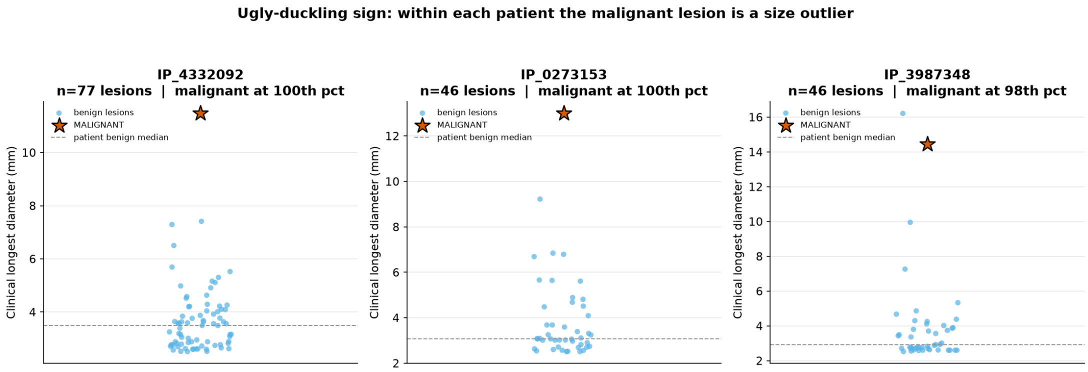

Figure 2: The ugly-duckling sign: the malignant lesion at the top of its own patient's size distribution.

5 Methods

Tabular FE (`src/features.py`): geometry/size ratios, color/hue/luminance contrasts, border/shape composites, 3D position, and the decisive **patient-relative ugly-duckling** deviations (`pdev_` / `prank_` / `pxc_`), plus per-fold smoothed target encoding — all computed fold-locally for leak-safety.

GBDT (`src/gbdt.py`): an early underfit bug (`is_unbalance` + AUC early-stop ⟶ `best_iter=1`, pAUC 0.099) was fixed by dropping `is_unbalance` and early-stopping on the official pAUC. The production expert bags 5-seed LightGBM with manual ~1% undersampling and rank-blends CatBoost (weight 0.2) ⟶ **OOF pAUC 0.16890** at near-zero cost.

Image experts (`src/vision/*`): small natural-image-pretrained backbones at 128/224 px, with per-epoch negative undersampling to ~1:1 (≈4.5 s/epoch; full 5-fold ≈ 8 min), BCE + label smoothing 0.05, `transV2` augmentation, EMA(0.995) + mixup. A 12-backbone sweep draws the frontier; heavy transformers collapse at 393 positives.

Stacking (`src/stack.py`): a **rank-average** of the GBDT and image OOFs — param-free and robust — beats a meta-LightGBM stacker, a learned per-lesion gate, and the addition of image embeddings.

A principled future direction is **self-supervised pretraining** (MAE / DINOv2) on the 400k+ unlabeled crops, as the in-constraint substitute for the banned external data.

6 Results

All numbers are out-of-fold, scored only by `src/cv.py` at 80% TPR, and independently re-derived from the OOF parquets by the `cv-guardian`.

Model	OOF pAUC@80%TPR	Params (M)	GFLOPs	CPU ms/img
GBDT (tabular only)	0.16890	0.86	~0	0.02
Best image (convnextv2_nano@224)	0.15945	14.98	2.46	60.9
Stack rank-avg[GBDT, nano@224]	**0.17430**	15.84	2.46	60.9
Cheap stack (gbdt+3img+udk)	0.17117	23.84	1.22	32.9
effvit_b0 (cheap frontier)	0.13706	2.13	0.034	3.5
mnv4_small (latency floor)	0.11242	2.49	0.062	3.1

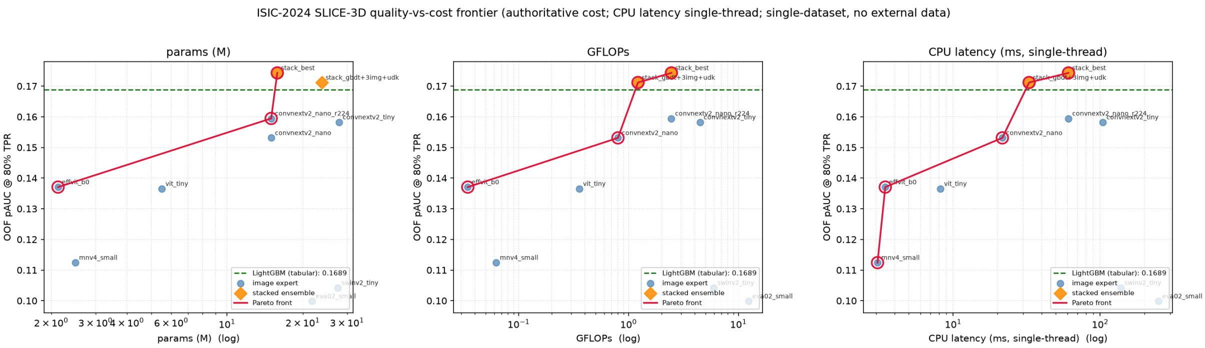

Figure 3: pAUC@80%TPR vs cost — the headline three-axis frontier.

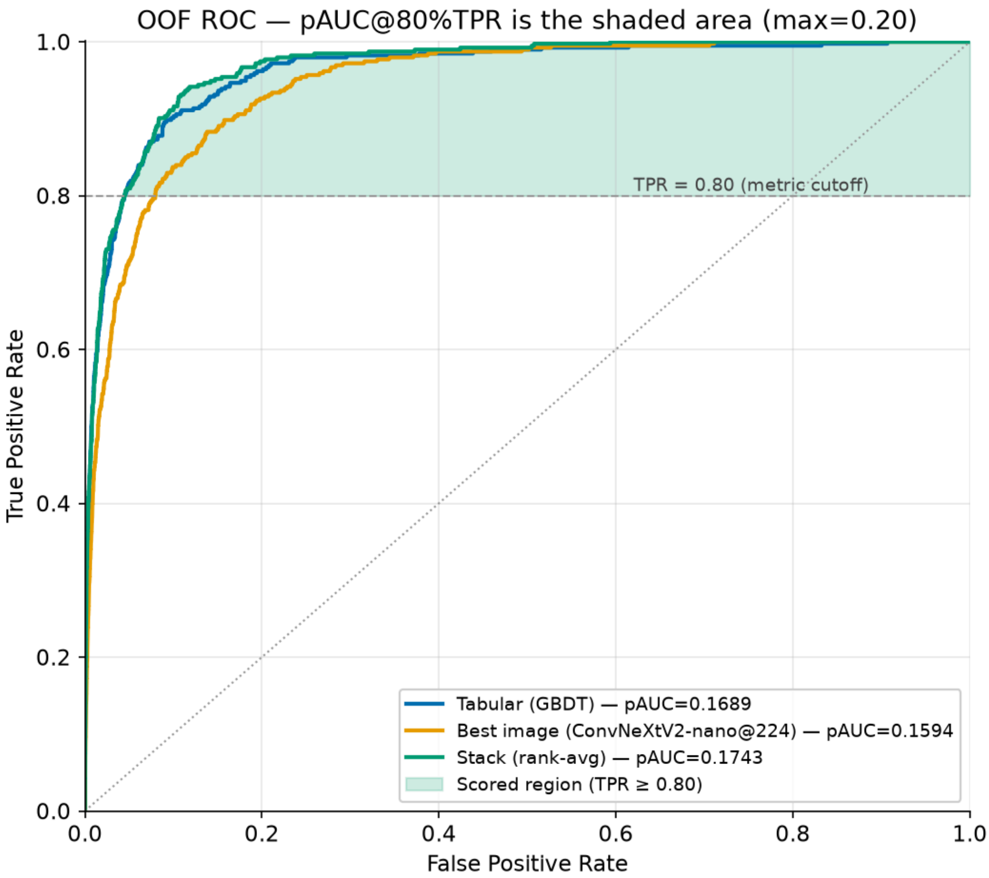

Figure 4: OOF ROC with the scored pAUC@80%TPR band shaded.

Per-fold: tabular 0.1693 ± 0.0112, image 0.1597 ± 0.0085, **stack 0.1749 ± 0.0068** — the stack is both the most accurate and the most stable (wins/ties on every fold). Patient-relative features account for ~65% of total GBDT gain. A fair private-test projection is the OOF headline minus ~0.01–0.02.

7 Ablations and negative results

Track	Progression / finding	pAUC
Tabular	broken ⟶ fixed ⟶ +ugly-duckling ⟶ bagged+CatBoost	0.099 ⟶ 0.118 ⟶ 0.144 ⟶ **0.169**
Image	nano@128 ⟶ nano@224; heavy backbones collapse	0.153 ⟶ **0.159**; Swin/EVA 0.104/0.100
Stack	rank-avg **wins** vs meta-LGBM vs gate vs +embeddings	**0.1743** vs 0.171 vs 0.150 vs <0.174

Negative results — complexity did not pay off (all OOF, leak-verified)

Idea tried	Its pAUC	Beaten by	Honest finding
Learned per-lesion gate (MoE)	0.15007	0.16890 (tabular)	LOSES to tabular alone
Meta-LGBM stacker	0.17108	0.17430 (rank-avg)	loses to trivial rank-avg
+ PCA image embeddings into GBDT	< 0.1743	0.17430	embeddings HURT
Heavy backbones (swinv2 / eva02)	0.104 / 0.100	0.15945 (nano@224)	dominated — overfit @393 pos
EMA 0.999 (over-smoothing)	~0.146	0.15945 (EMA0.995)	stronger EMA hurts

Figure 5: Negative-results summary: every complexity increase lost.

The structural reason: every learned add-on must fit its parameters from the same 393 positives the base experts already used; a zero-parameter rank-average cannot overfit. **Complexity consistently lost — that is the contribution.**

8 Reproducibility

Conda env `isic2024` (Python 3.12, torch 2.11.0+cu128), `SEED = 42` everywhere, frozen `data/folds.parquet`, single RTX 4070 Ti SUPER 16 GB. Commands: `make folds` (always first) ⟶ `make gbdt` ⟶ `make vision CFG=...` ⟶ `make frontier` ⟶ `make test` (9/9) ⟶ `make site`. The `cv-guardian` reproduced all headline numbers from disk.

9 Credits and positioning

We credit the leaderboard champions — 1st place **Ilya Novoselskiy** (EVA-02 + EdgeNeXt + GBDT, private pAUC 0.17264; used external ISIC data + ~30k synthetic lesions, both banned here; their own ablation shows synthetic added only +0.0007), 2nd **uchiyama33**, 3rd **kyohei-123** — and the public-notebook lineage we learned from (greysky's tabular recipe, snnclsr, motono0223, andreasbis's "CNN preds as features", richolson). Our own contributions are the efficiency frontier as the deliverable, the audited no-external/no-synthetic discipline, the image-space ugly-duckling, and the honest negative-results catalogue. We claim SOTA *within the no-external/no-synthetic class*, not against the unconstrained ~0.1755.

10 References

Bibliography